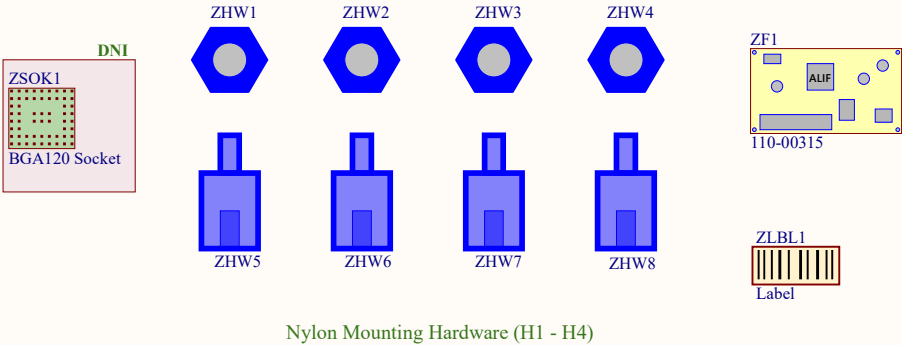


Revision History:

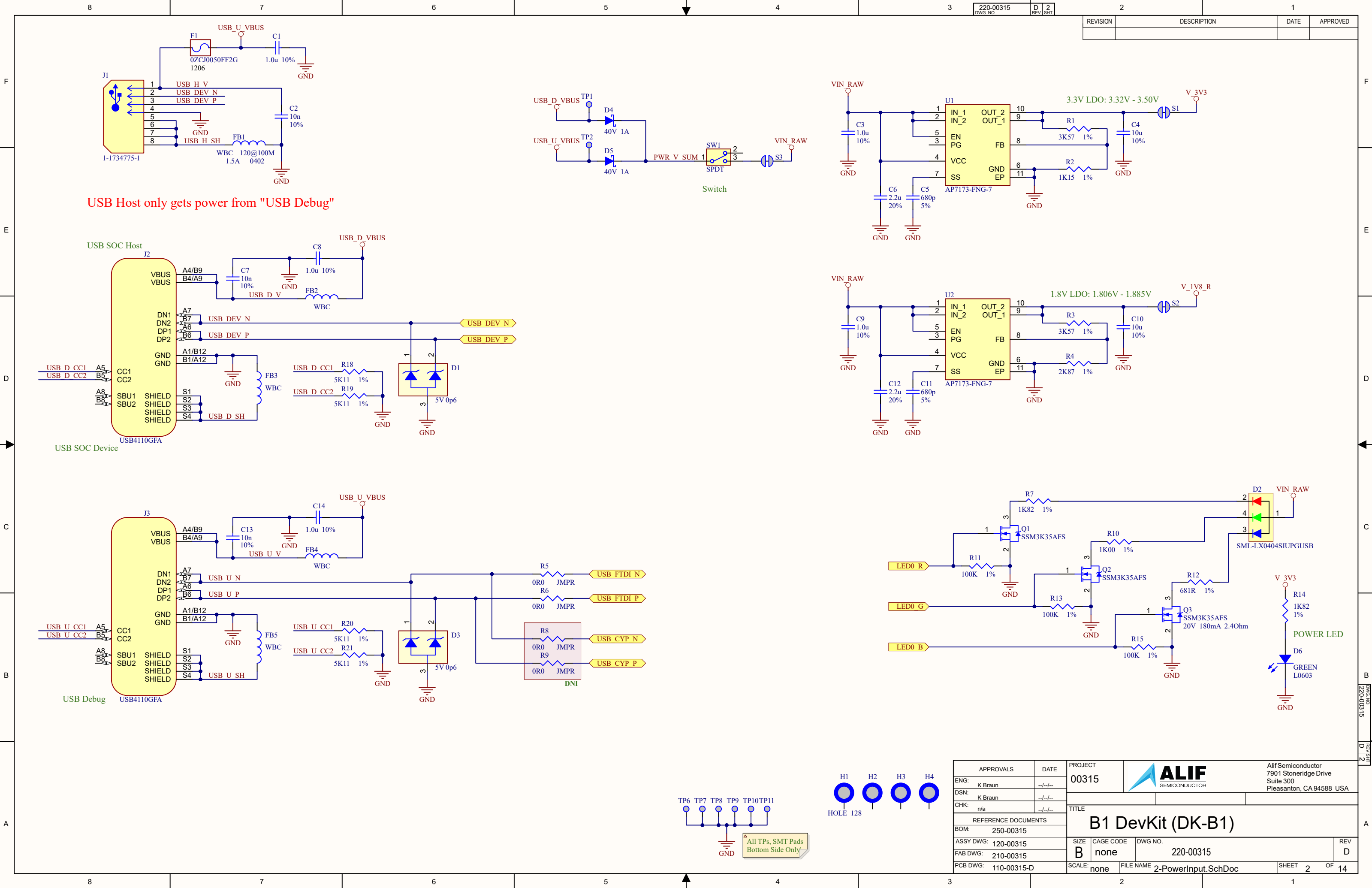
REV	DATE	Description:
A	16-APR-2024	Initial Release
A1	11-JUL-2024	I3C pullups need to connect ro V_1V8_R. Add wire
	12-JUL-2024	Click UART RX and TX are swapped
		- Not easily reworkable
B	22-JUL-2024	Labeling LPCAM_VSYNC and HSYNC on J14 is swapped, GPIO connections are ok
	24-JUL-2024	Will need to modify Solder Mask & Paste Layer on U3
	26-JUL-2024	CAN UART RXD and TXD are swapped at U15.
		Incorporate all changes mentioned in A1
	06-AUG-2024	Added shunt S4 to break CAN_RXD from U15
no B1 or B2		Changed JLink OB back to Ensemble E1 BGA120 device
		Added 1M ohm bleeder resistor to VREG_LP_1V8 to address startup issue
		Replaced OSPI Combo device with OSPI Flash and HyperRAM used in Ensemble DevKit / AppKit
B3	27-AUG-2024	Replaced RF Switch J9 with a U.FL compatable device
		Changed Antenna matching components
	19-NOV-2024	Changed BLE RF Front End for LPA operation
C	22-NOV-2024	Convert to JLink OB
	08-APR-2025	Alif A5 BGA ball change, added 2 external BOR devices
	14-MAY-2025	Added shunts for current measurement jumpers and DNI'd JP1-JP7
C1		Dimmed JLink-OB green LED
		With A5, Port 9 is gone:
		Deleting I2S Mics
D		Merged and shared Click I2C0 into I2C1
		Moved LPUART_TX
		Added Power Enable for SD Micro
D		Remove J4 "Power Only" USB-C
		Alif A6 power decoupling updates C17, C23, C86, FB8, FB12.
		Remove FB9, C24
D		Adding R7 in parallel with FB12. Removed U4, precision VREF

Required 2.54mm Shunts:

JP1	VDD_3V3 USB Current Measurement (* shunted)
JP2	VDD_3V3 BUCK Current Measurement (* shunted)
JP3-1	VDD_SOC = 3.3V
JP3-2	VDD_SOC = 1.8V (* shunted)
JP4	VDD_3V3 MAIN Current Measurement (* shunted)
JP5	VDDIO_1V8 Current Measurement (* shunted)
JP6-1	FLEXIO = 3.3V
JP6-2	FLEXIO = 1.8V (* shunted)
JP7-1	VREG_DIG_1V8 = 1uF + 10ohm
JP7-2	VREG_DIG_1V8 = 100nF (* shunted)
Click Board Source Voltage:	
JP9-1	Click Board Power = 1.8V
JP9-3	Click Board Power = 3.3V (* shunted)
JP9-5	Click Board Power = 5.0V
JP10	Hold JLink OB E1 in reset (for JTAG)



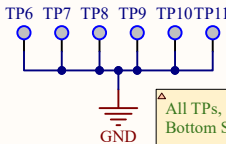
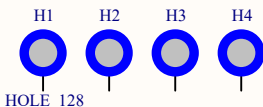
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ENG:	K Braun	--/--/--	00315					
DSN:	K Braun	--/--/--	TITLE					
CHK:	n/a	--/--/--						
REFERENCE DOCUMENTS			B1 DevKit (DK-B1)					
BOM: 250-00315								
ASSY DWG:	120-00315	SIZE				CAGE CODE	DWG NO.	REV
FAB DWG:	210-00315	B				none	220-00315	D
PCB DWG:	110-00315-D	SCALE:				none	FILE NAME	1-RevisionHistory.SchDoc



USB Host only gets power from "USB Debug"

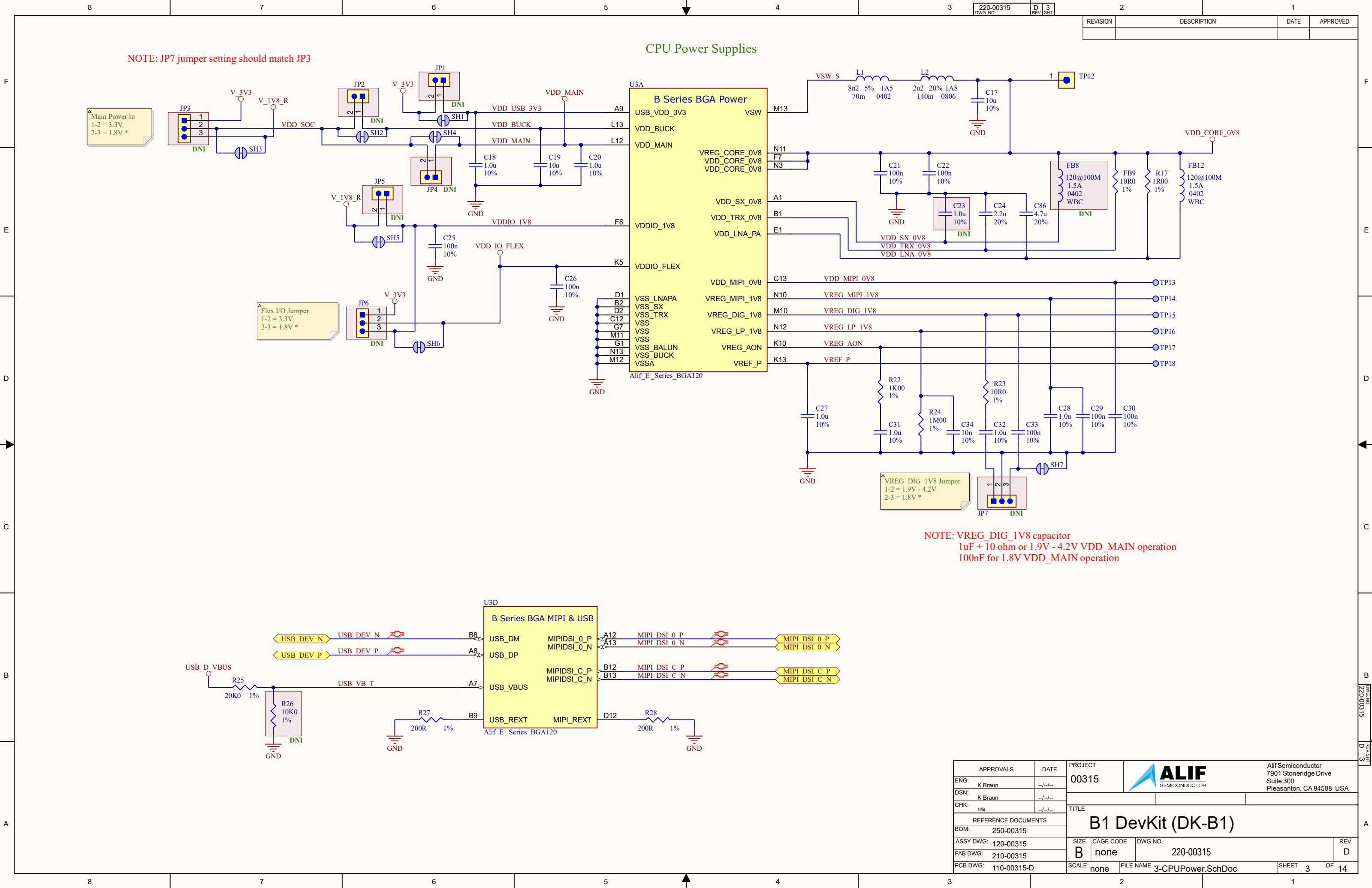
Switch

REVISION	DESCRIPTION	DATE	APPROVED



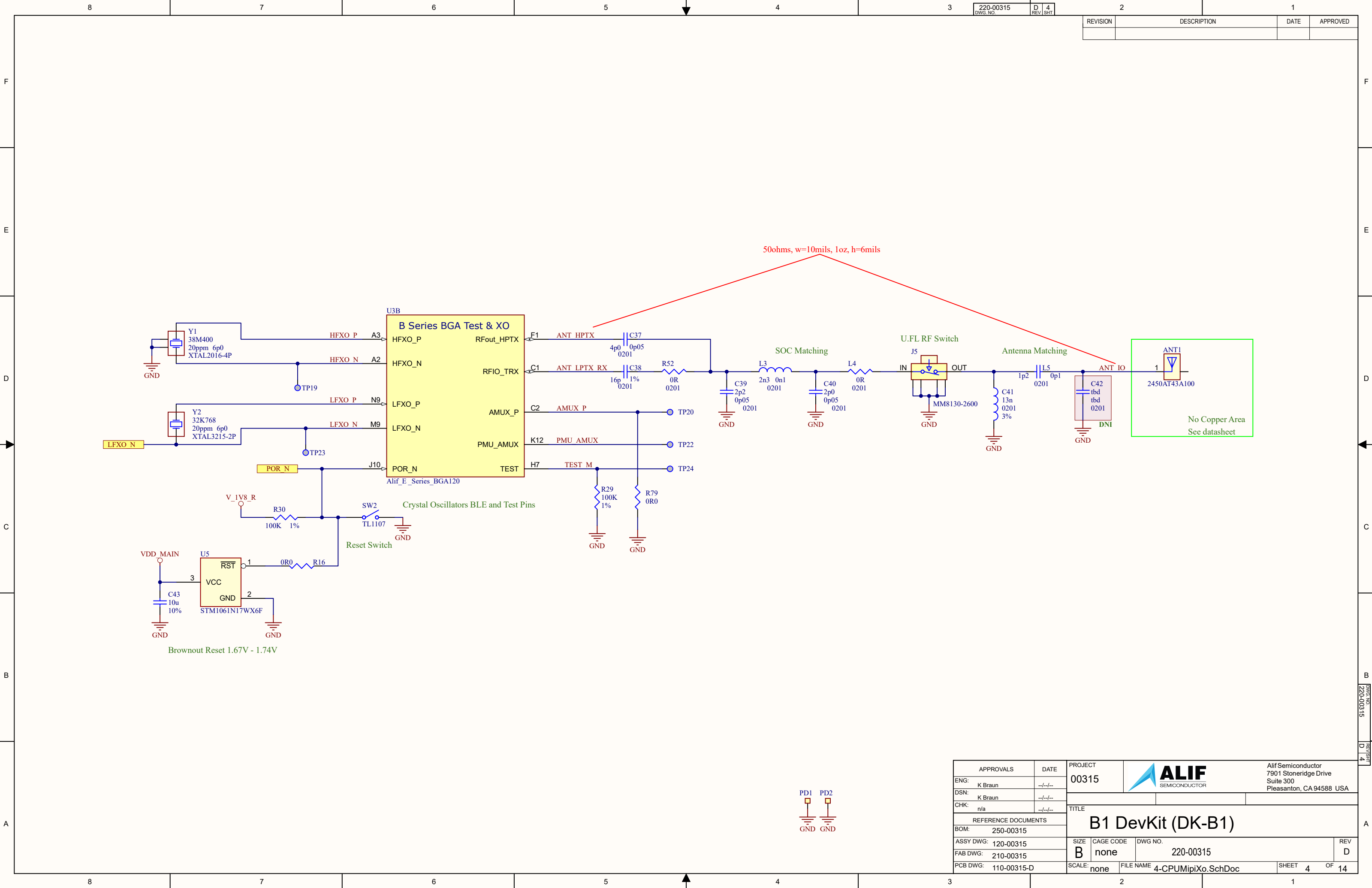
All TPs, SMT Pads
Bottom Side Only!

APPROVALS	DATE	PROJECT	220-00315	
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DSN: K Braun	--/--		Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA	
CHK: n/a	--/--			
REFERENCE DOCUMENTS		TITLE		
BOM: 250-00315		B1 DevKit (DK-B1)		
ASSY DWG: 120-00315	SIZE B	CAGE CODE none	DWG NO. 220-00315	REV D
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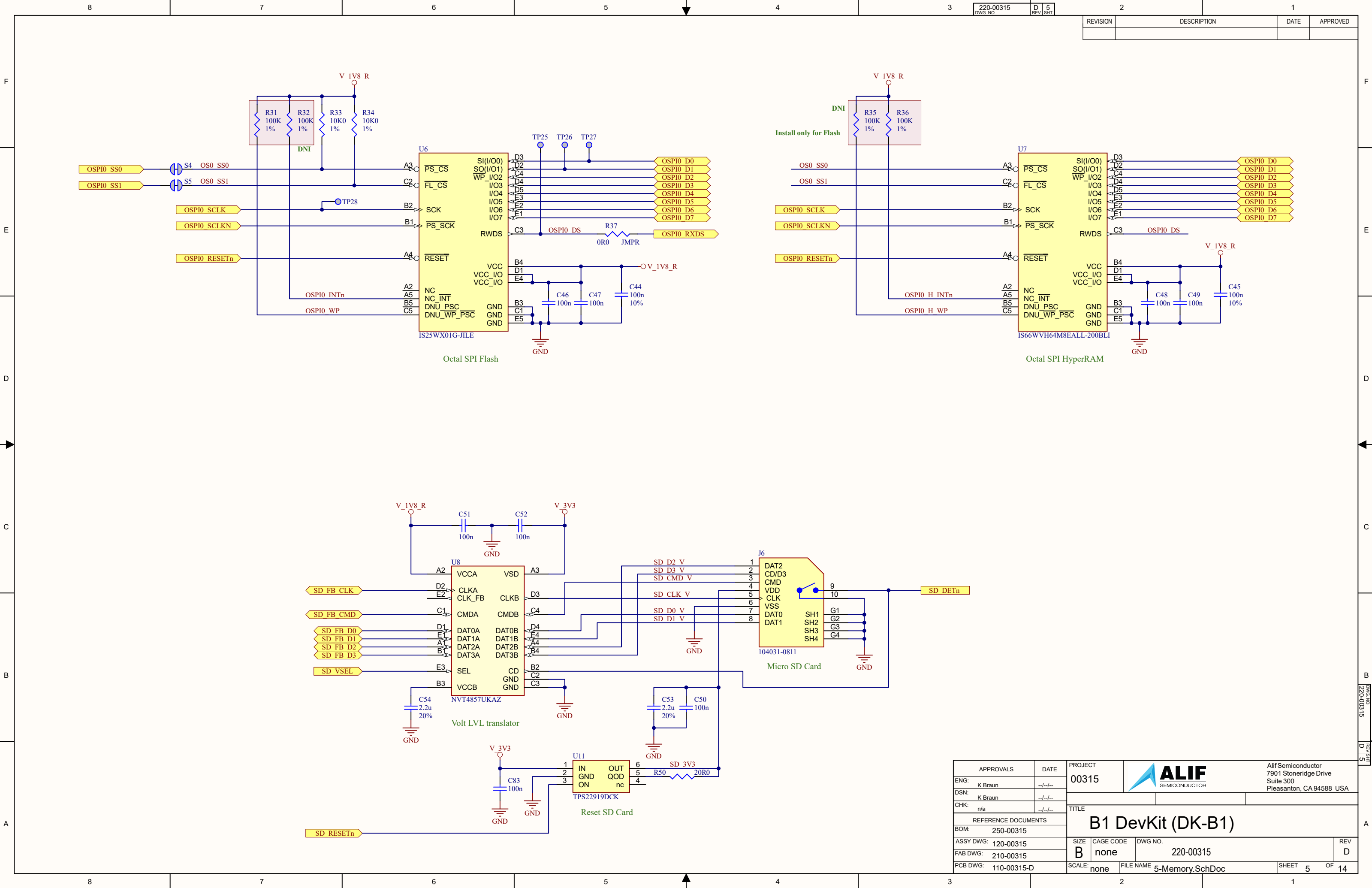


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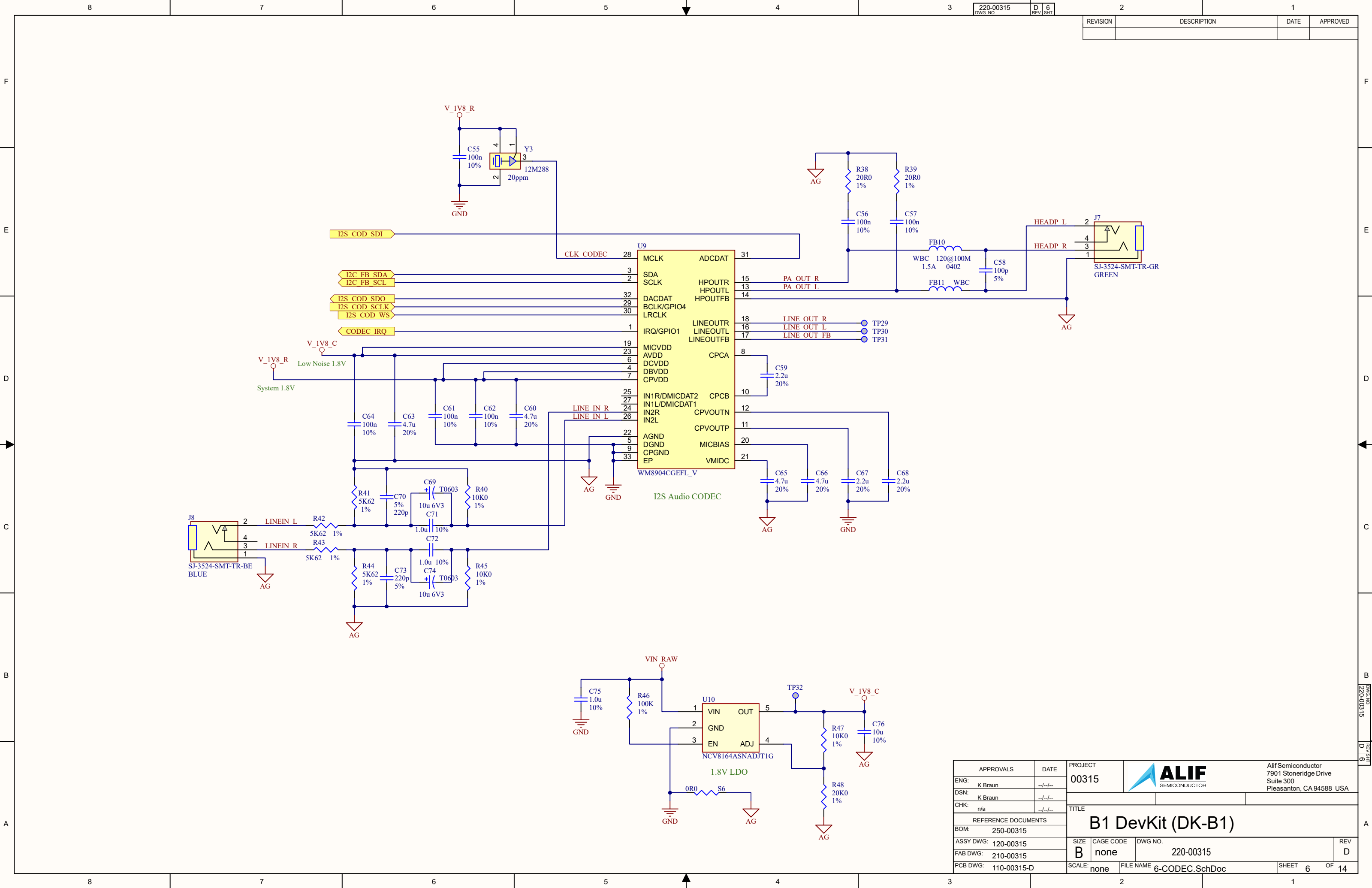
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APPROVALS		DATE	PROJECT		 ALIF SEMICONDUCTOR Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA	
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CHK:	n/a	--/--/--				
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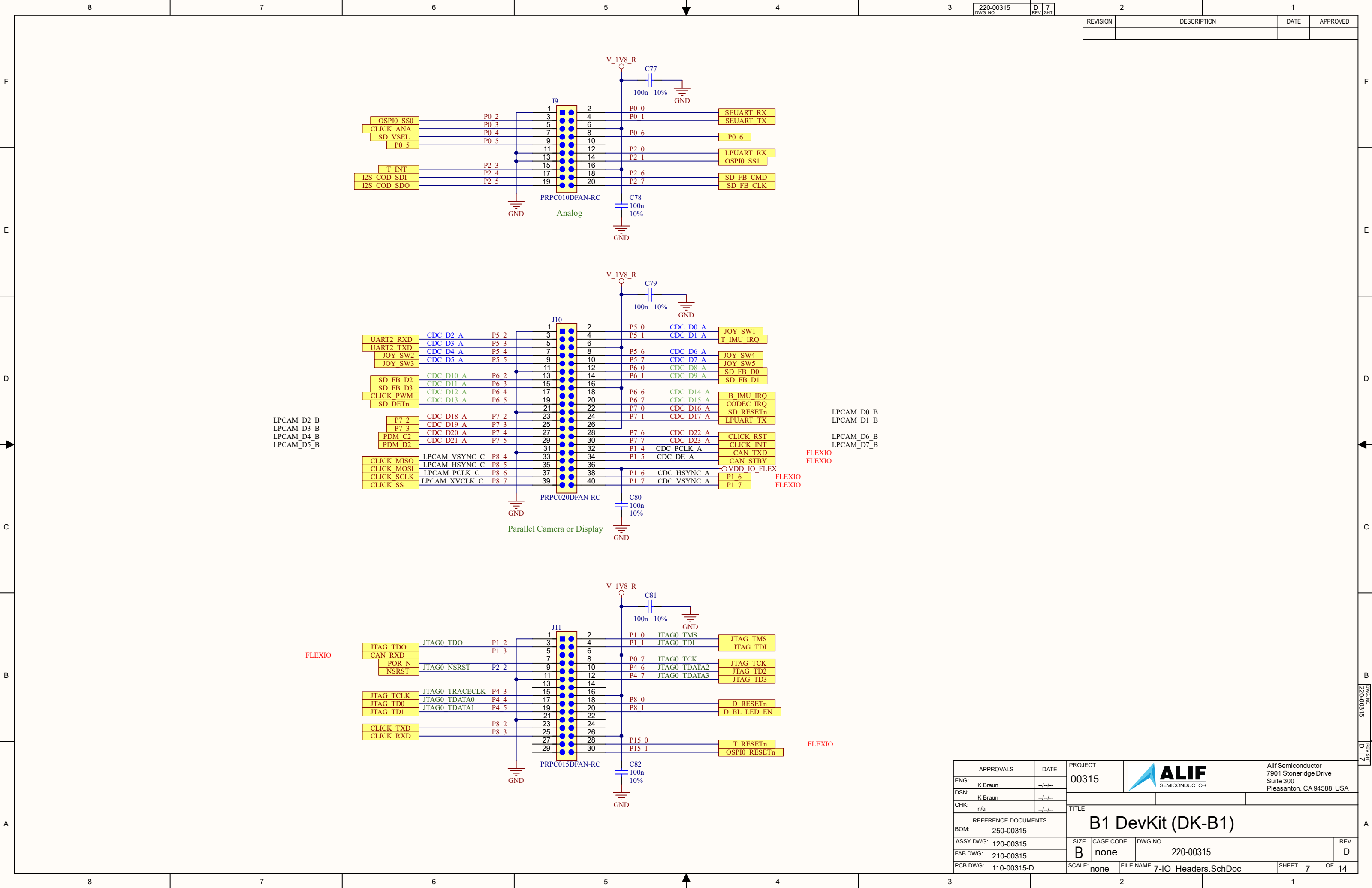


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CHK: n/a		--/--							
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PCB DWG: 110-00315-D									



REVISION	DESCRIPTION	DATE	APPROVED

APPROVALS		DATE	PROJECT		 ALIF SEMICONDUCTOR		Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA	
ENG:	K Braun	--/--/--	00315					
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PCB DWG:	110-00315-D		SCALE:	none	FILE NAME 6-CODEC.SchDoc		SHEET 6 OF 14	



REVISION	DESCRIPTION	DATE	APPROVED

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DSN:	K Braun	--/--/--								
CHK:	n/a	--/--/--								
REFERENCE DOCUMENTS			B1 DevKit (DK-B1)							
BOM:	250-00315									
ASSY DWG:	120-00315									
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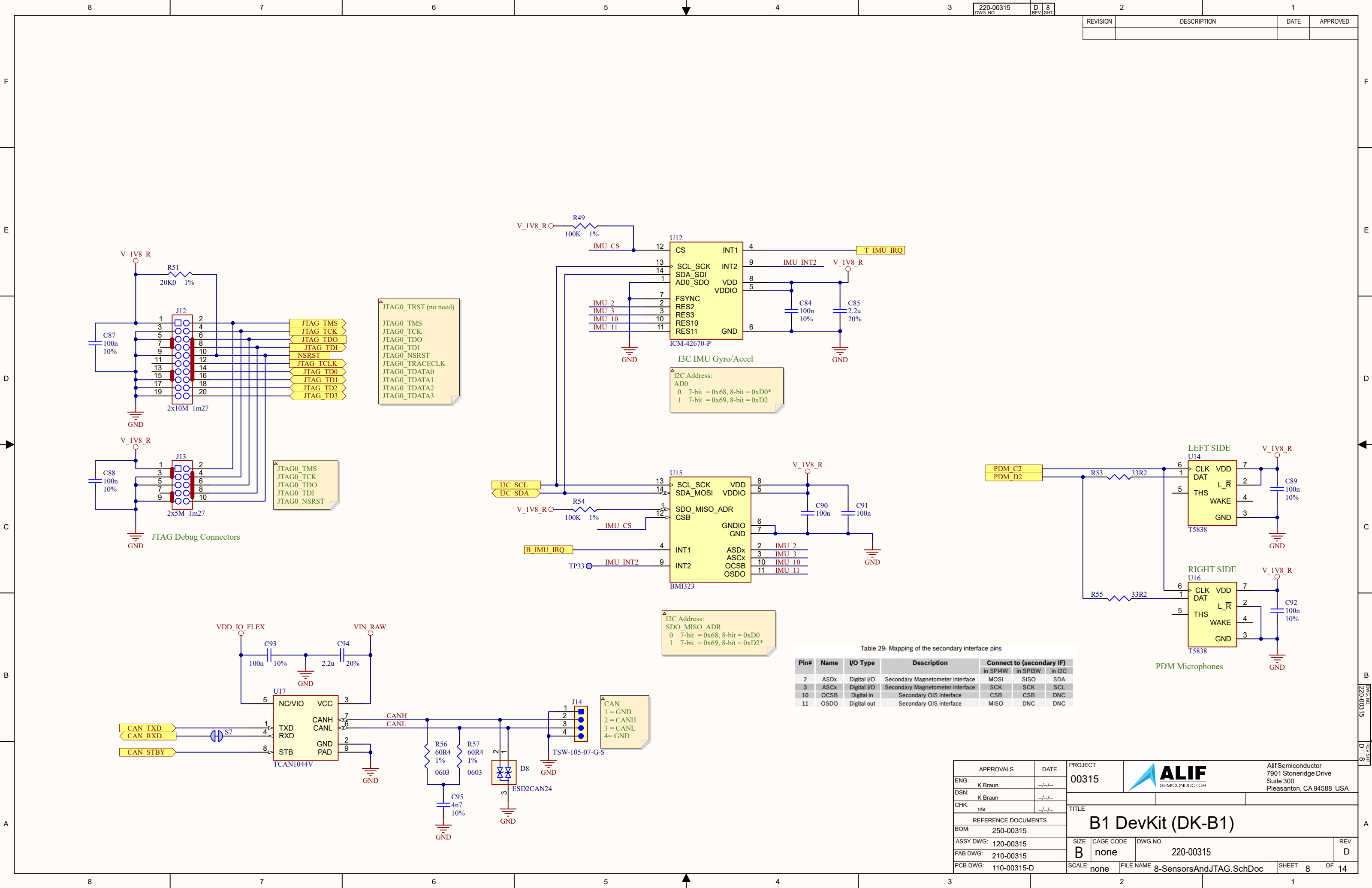
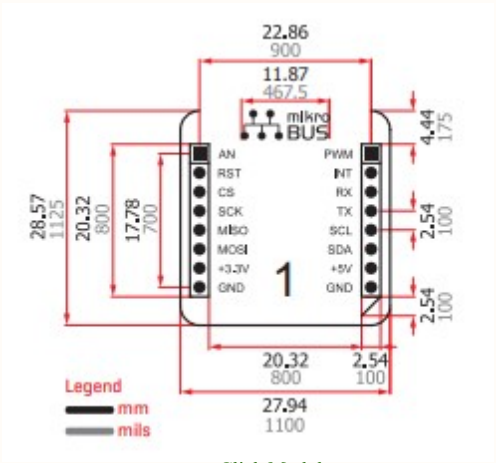
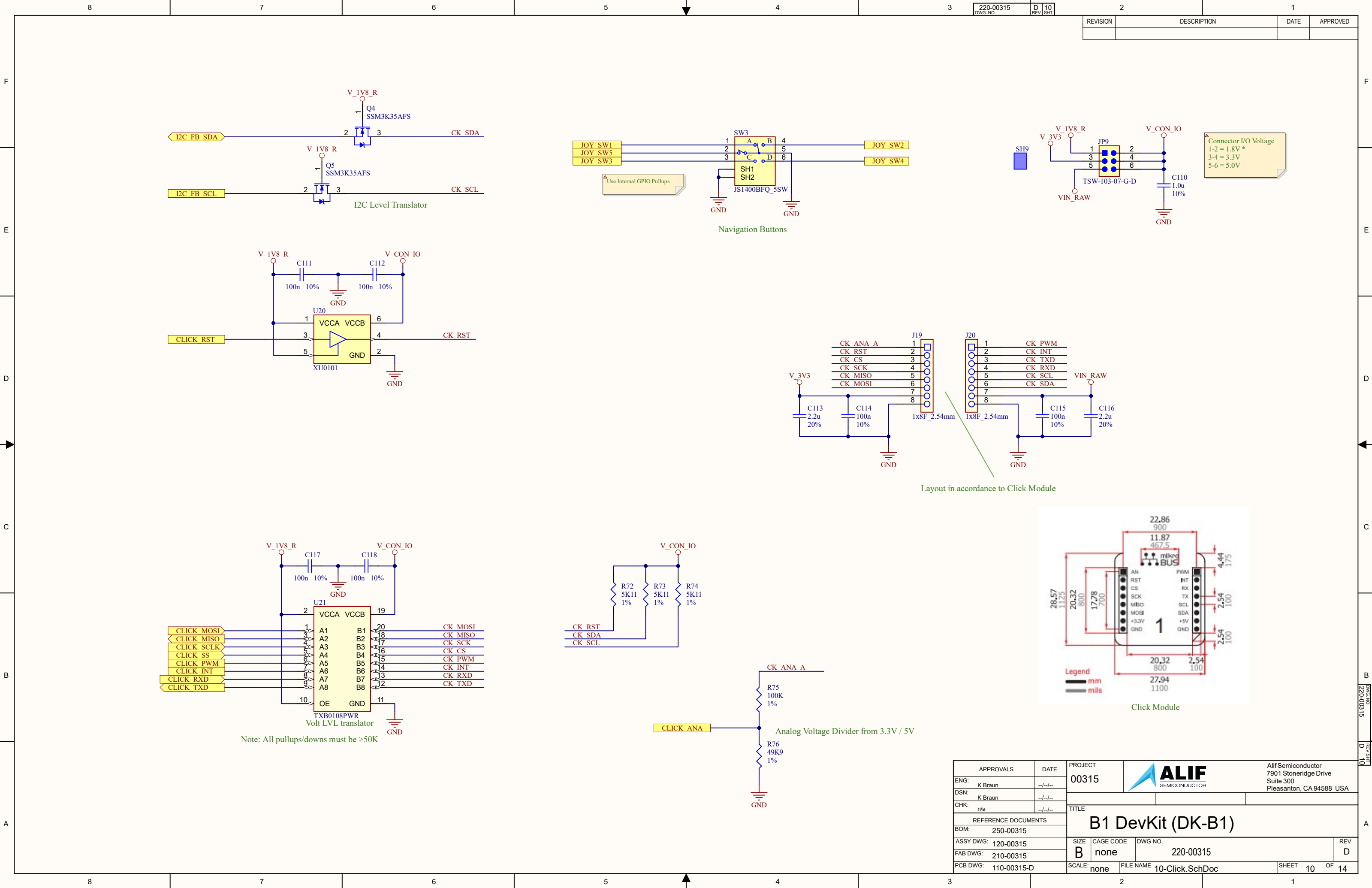
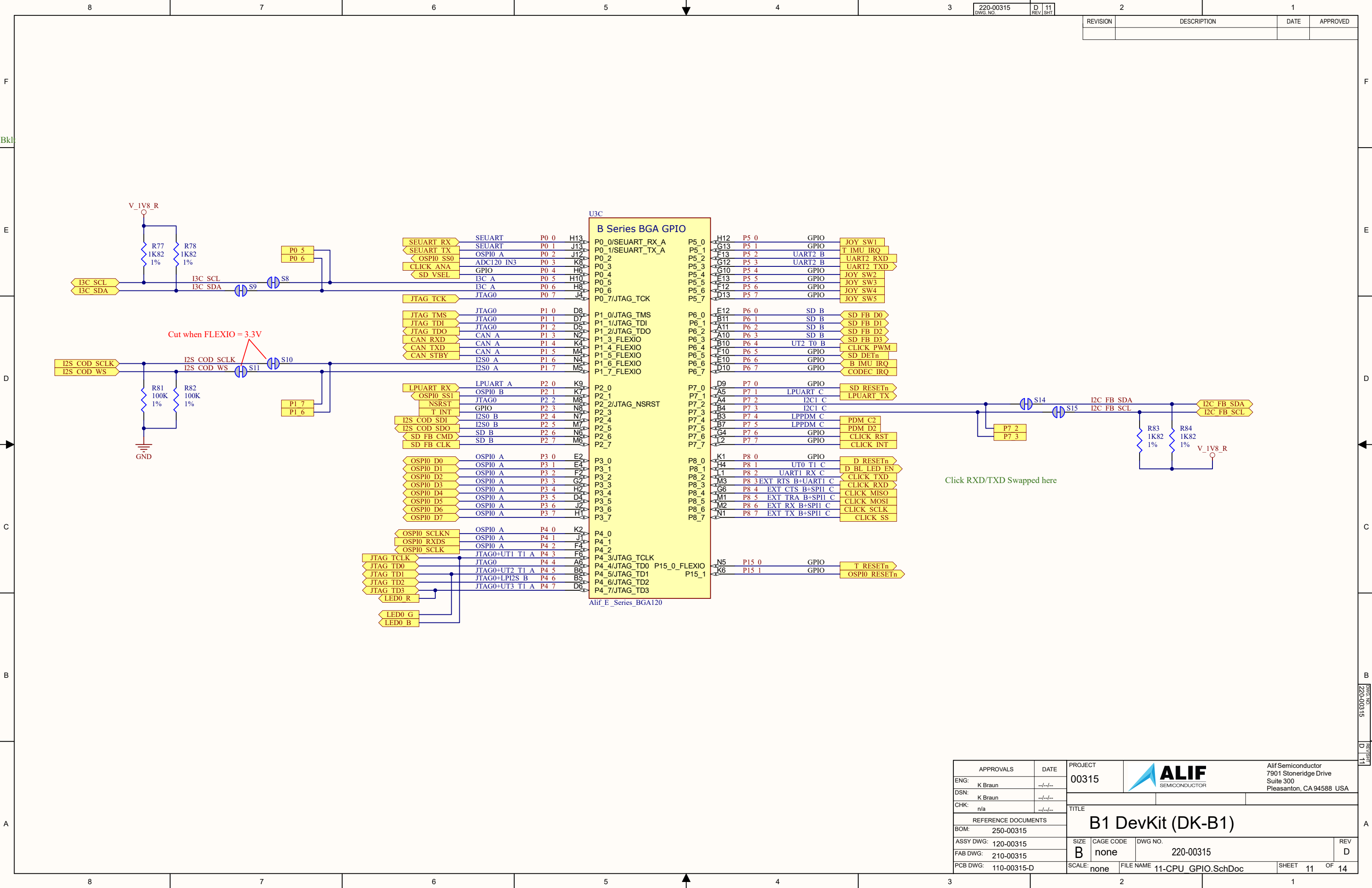


Table 29: Mapping of the secondary interface pins						
Pin#	Name	I/O Type	Description	Connect to (secondary IF)		
				in SPI4W	in SPI3W	in I2C
2	ASDx	Digital I/O	Secondary Magnetometer interface	MOSI	SISO	SDA
3	ASCx	Digital I/O	Secondary Magnetometer interface	SCK	SCK	SCL
10	OCSB	Digital in	Secondary OIS interface	CSB	CSB	DNC
11	OSDO	Digital out	Secondary OIS interface	MISO	DNC	DNC

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ENG:	K Braun	--/--/--	00315						
DSN:	K Braun	--/--/--							
CHK:	n/a	--/--/--							
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BOM:	250-00315								
ASSY DWG: 120-00315			SIZE	CAGE CODE	DWG NO.		REV		
FAB DWG: 210-00315			B	none	220-00315		D		
PCB DWG: 110-00315-D			SCALE:	none	FILE NAME	8-SensorsAndJTAG.SchDoc		SHEET 8 OF 14	

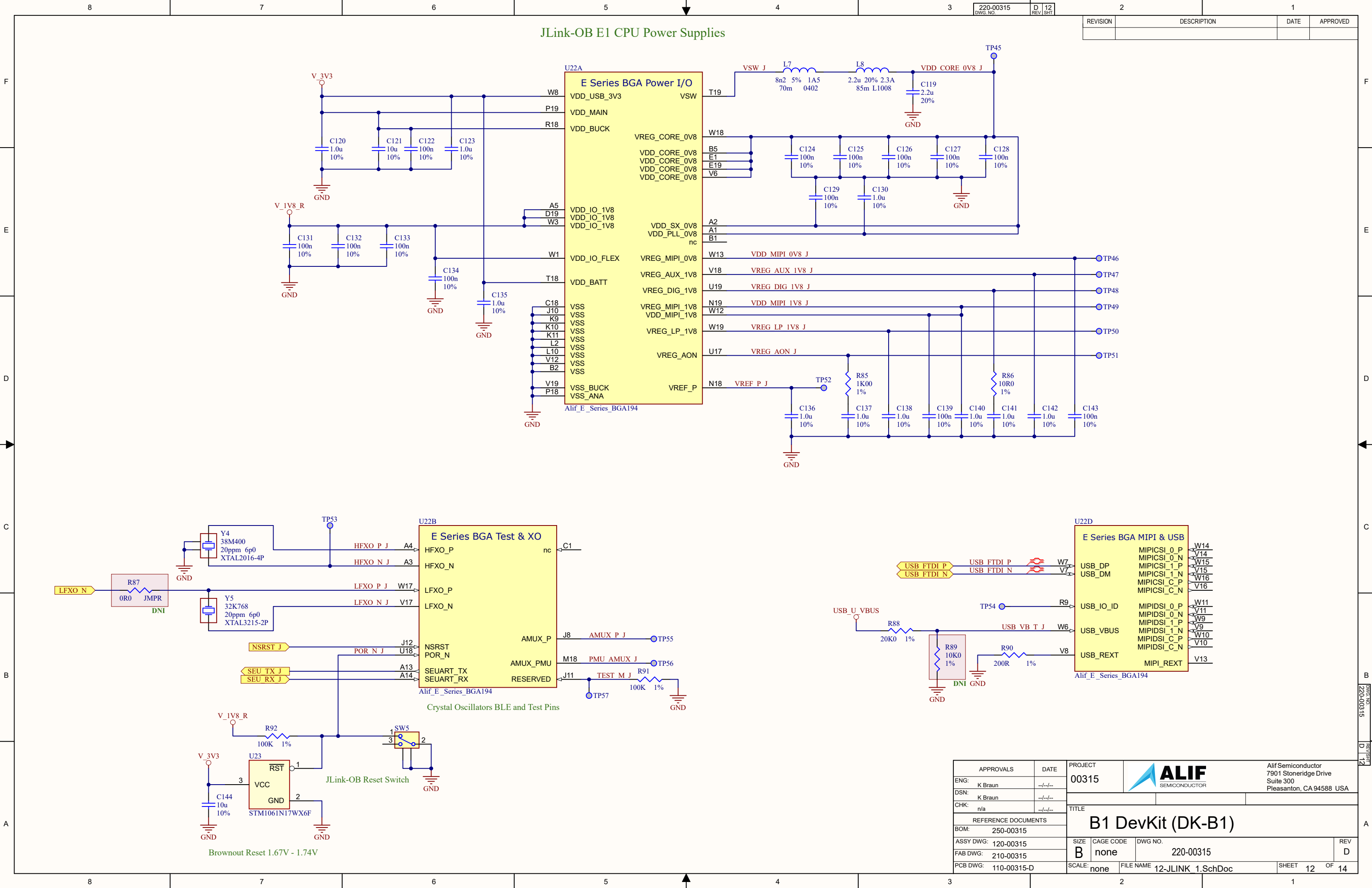


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BOM: 250-00315								
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FAB DWG: 210-00315			B	none	220-00315		D	
PCB DWG: 110-00315-D			SCALE:	FILE NAME	SHEET		OF	
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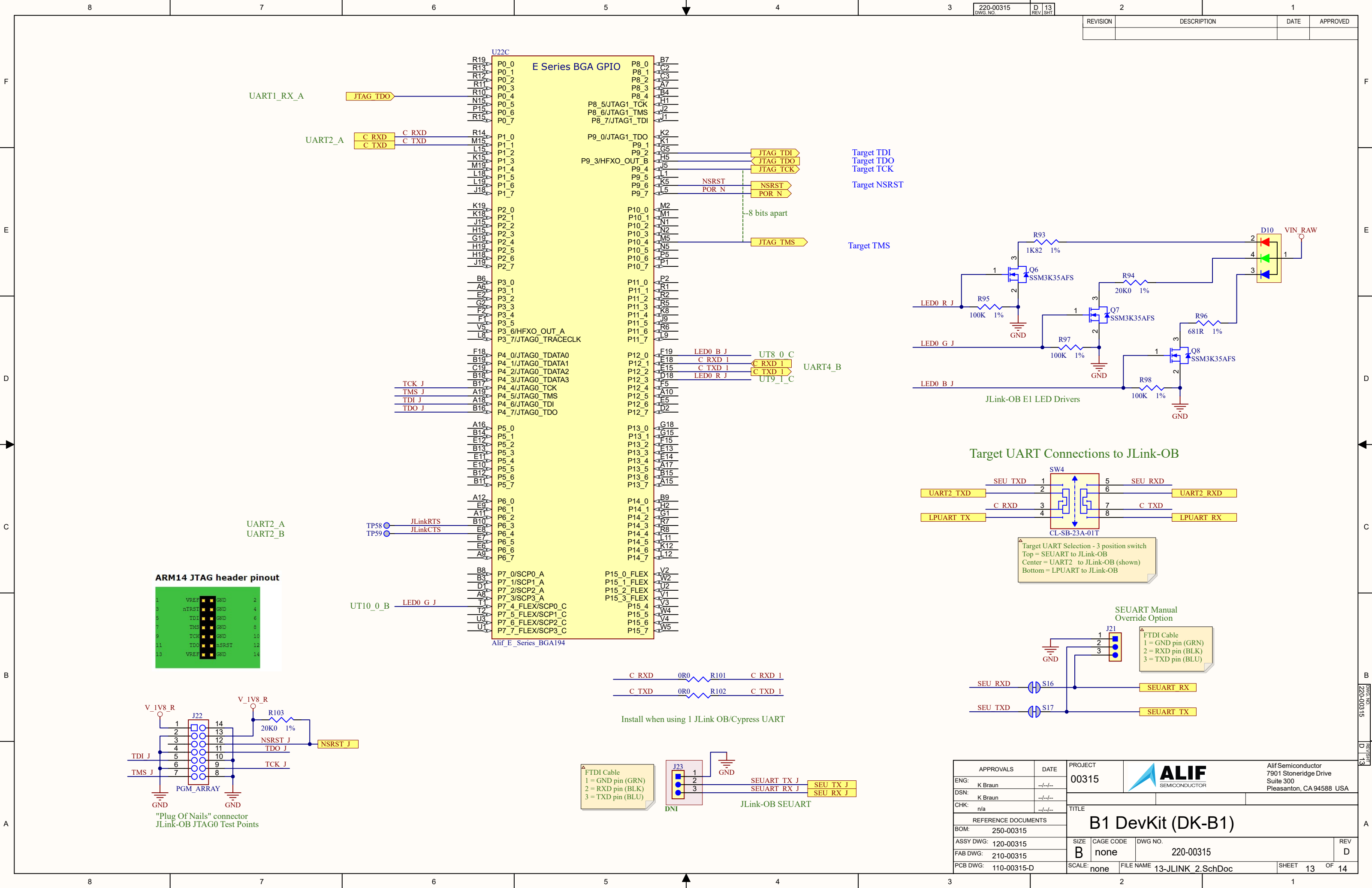


REVISION	DESCRIPTION	DATE	APPROVED

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ENG:	K Braun	--/--/--	00315								
DSN:	K Braun	--/--/--			TITLE						
CHK:	n/a	--/--/--									
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BOM:	250-00315										
ASSY DWG:	120-00315										
FAB DWG:	210-00315										
PCB DWG:	110-00315-D		SCALE:	none	FILE NAME	11-CPU GPIO.SchDoc		SHEET	11	OF	14



APPROVALS		DATE		PROJECT		 ALIF SEMICONDUCTOR		Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA					
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CHK: n/a		--/--/--											
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ASSY DWG: 120-00315													
FAB DWG: 210-00315													
PCB DWG: 110-00315-D				SCALE:	none	FILE NAME	12-JLINK 1.SchDoc			SHEET	12	OF	14



REVISION	DESCRIPTION	DATE	APPROVED

APPROVALS		DATE	PROJECT		ALIF SEMICONDUCTOR Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA	
ENG:	K Braun	--/--/--	00315			
DSN:	K Braun	--/--/--	TITLE			
CHK:	n/a	--/--/--				
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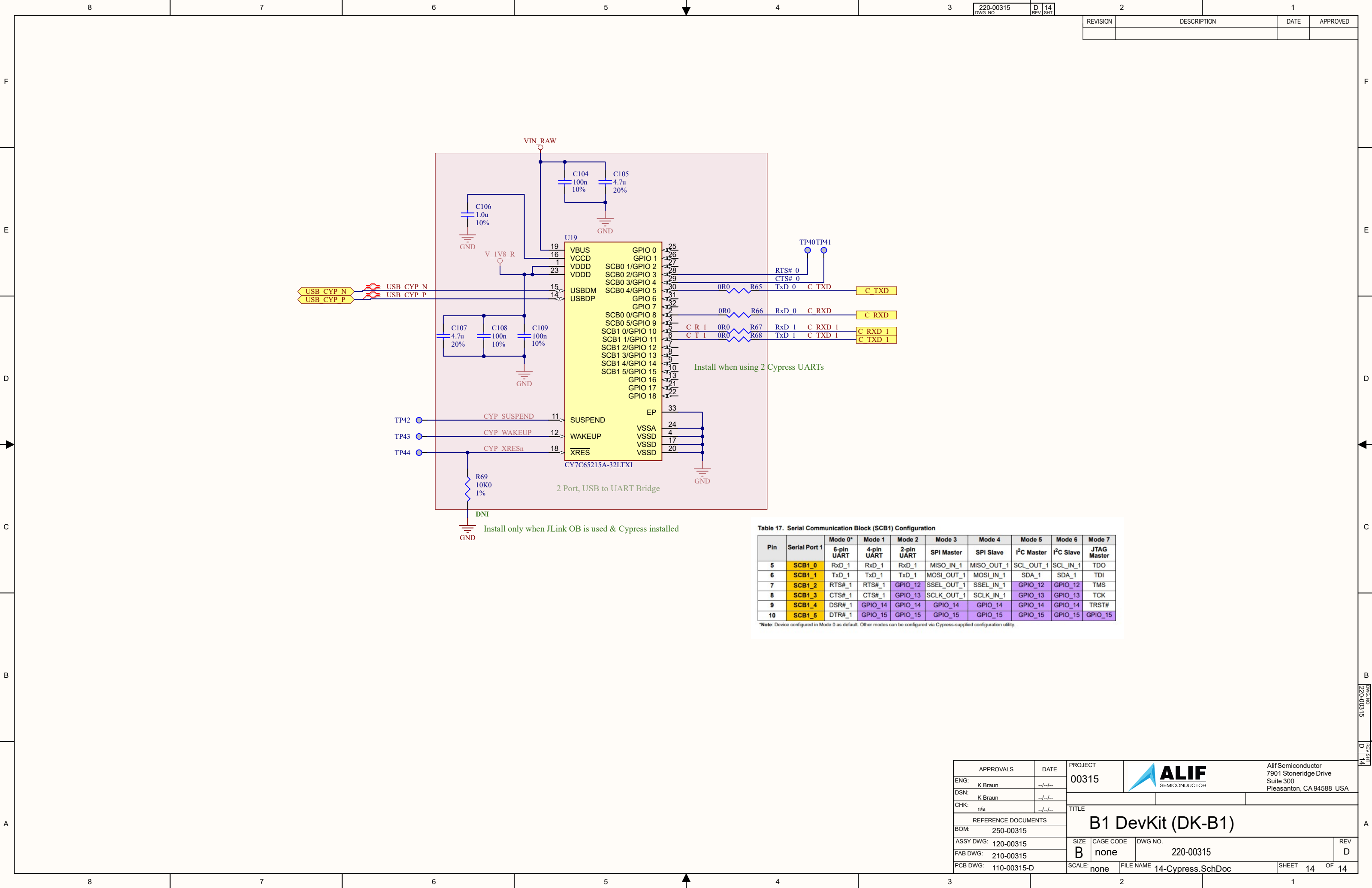


Table 17. Serial Communication Block (SCB1) Configuration

Pin	Serial Port 1	Mode 0*	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7
		6-pin UART	4-pin UART	2-pin UART	SPI Master	SPI Slave	I ² C Master	I ² C Slave	JTAG Master
5	SCB1_0	RxD_1	RxD_1	RxD_1	MISO_IN_1	MISO_OUT_1	SCL_OUT_1	SCL_IN_1	TDO
6	SCB1_1	TxD_1	TxD_1	TxD_1	MOSI_OUT_1	MOSI_IN_1	SDA_1	SDA_1	TDI
7	SCB1_2	RTS#_1	RTS#_1	GPIO_12	SSEL_OUT_1	SSEL_IN_1	GPIO_12	GPIO_12	TMS
8	SCB1_3	CTS#_1	CTS#_1	GPIO_13	SCLK_OUT_1	SCLK_IN_1	GPIO_13	GPIO_13	TCK
9	SCB1_4	DSR#_1	GPIO_14	GPIO_14	GPIO_14	GPIO_14	GPIO_14	GPIO_14	TRST#
10	SCB1_5	DTR#_1	GPIO_15	GPIO_15	GPIO_15	GPIO_15	GPIO_15	GPIO_15	GPIO_15

*Note: Device configured in Mode 0 as default. Other modes can be configured via Cypress-supplied configuration utility.

APPROVALS		DATE	PROJECT		 ALIF SEMICONDUCTOR		Alif Semiconductor 7901 Stoneridge Drive Suite 300 Pleasanton, CA 94588 USA		
ENG: K Braun		--/--/--	00315						
DSN: K Braun		--/--/--							
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REFERENCE DOCUMENTS				TITLE					
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ASSY DWG: 120-00315									
FAB DWG: 210-00315				SIZE B	CAGE CODE none	DWG NO. 220-00315	REV D		
PCB DWG: 110-00315-D				SCALE: none	FILE NAME 14-Cypress.SchDoc	SHEET 14 OF 14			